The Problems with Accepting An Invitation to Cognitive Science

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For experienced cognitive scientists with a philosophical bent who want to know more about the history of attempts to apply logic to modeling cognition, Justin Leiber's *Invitation to Cognitive Science* will be a delight. However, if you are not already well-versed in the multidisciplinary field of cognitive science, you probably should not accept this invitation. This is a book written mostly for philosophers and, as such, it will be of limited value to someone looking for an introduction to cognitive science.

In the early chapters of the book, Leiber takes us on an epistemological journey from Socrates to the present, focusing largely on the work of Frege, Russell, early Wittgenstein and Turing. He is one of the few authors who gives La Mettrie his rightful place in the pantheon of those who have believed in the idea of mind as machine. He also treats us to a nice discussion of the ideas of Wittgenstein and Turing. His presentation of Turing is at times fascinating, including many little-known aspects of his life and work. Especially interesting is the reporting of the discussions between Turing and Wittgenstein. Overall, however, the book attaches too much importance to logic in cognition. Leiber claims, for example, "Russell's work made clear the centrality and power of logic for all cognitive study and, in doing so, suggested how it might both show us much about the content and the form of thought. . . . the model strongly suggests that the production of thoughts, of chains of reasoning -- *thinking* -- must be like theorem-proving." Granted, neuro-biological rules govern the lowest level of our brains, but one must be very careful not to confuse the logical character of these rules with the implication that there must exist laws of thought at the much higher cognitive levels. This notion that cognition is driven by an underlying (and ultimately, discoverable) "logic of thought" has been the basis, in one form or another, of traditional artificial intelligence since its inception. This belief has led to enormous problems -- in particular, problems of computational intractability and combinatorial explosion. While Leiber may allude to these problems on occasion, he does not stress them, even though they lie at the heart of modeling human cognition, cropping up in *every area* of human cognition.

The essence of cognitive science is its *multidisciplinary* nature, a fact that is not reflected in Leiber's treatment of the subject. Cognitive science is the study of the how numerous disciplines -- traditionally, cognitive psychology,

computer science, philosophy, neuroscience, and linguistics -- have approached the problem of understanding the mind. But this book focuses almost exclusively on the history of attempts to model human reasoning using logical formalisms and thereby tells only a small, if interesting, part of the story.

Leiber's approach to cognition is deeply rooted in the philosophies of Chomsky, Russell, Frege and early Wittgenstein, approaches in which formal logic rules supreme. Unfortunately, however, formal logic *does not* rule supreme in the realm of real cognition. While logic obviously has its place in cognition -- for example, when we try to discover why our car refuses to start or to debug a computer program -- it is by no means the whole story. In reality, logic accounts for only a small part of the broader picture of cognitive activity. Obstinate attempts to straightjacket cognition with predicate-calculus representations manipulated by the laws of Boole, Frege, and Russell, constitute one of the primary reasons that the progress in modeling cognition has faltered badly. This is certainly not an obvious point and for many years there have been two opposing camps in the field of cognitive science/artificial intelligence (see Hofstadter 1982d). One group believes in the ultimate success of "top-down" rule-based techniques to model cognition while the other has put their money on so-called "bottom-up" (and most recently, "connectionist") approaches to modeling cognition. These latter techniques explicitly repudiate high-level rules and formal logic. The truth almost certainly lies somewhere in between and, in all probability, the future of cognitive modeling lies with models that can integrate both implicit information as well as rules. These conflicting approaches and their potential resolution are hardly mentioned in Leiber's book.

For example, consider what it would take, in terms of the vast amount of interconnected knowledge of subtle associations, phonetics, cultural connotations, etc., for a computer program to be able to determine, as any native speaker of English can do instantly, that *Flugly* -- a made-up name -- would be a good name for an accountant in a W.C. Fields movie, a lousy name for a glamorous Hollywood actress, and a good name for a child's teddy bear? Or that calling someone a *trubhead* is not very nice. Or that the word *blutch* is not as pretty as *farfellatta*. Or imagine what it would take for a computer program to have a representational structure so rich, so malleable that it could understand (or, better, generate) the following joke: "Why is marriage like flies on a window pane?" (Answer: Because those on the outside want to get in and those on inside want to get out.) Or ask a hundred people to tell you the first answer that pops into their head when you ask them, "What do cows drink?" Getting computers to produce *that* answer (as well as the right answer) is what modeling human cognition is all about. Leiber devotes a whole chapter to the Turing Test for intelligence without seemingly realizing that perfectly ordinary kinds of questions, whose answers rely on our vast network of learned, but largely unconscious associations, would make it virtually impossible for any computer that had not experienced the world as we had to pass the Turing Test*. *These* issues --

issues involving representations, concepts, learning, generalization, emergence, similarity and sameness -- are the central issues of cognitive science. But these crucial areas -- especially crucial in an *introduction* to cognitive science are hardly, if ever, addressed in this book.

There is a tacit assumption in most rule-based approaches to cognitive modeling —namely, that cognition involves manipulating representations of reality by means of rules. Even if one were to accept this, it leaves unanswered the following all-important question: *Where do the representations to be manipulated come from*? Leiber touches on this briefly when discussing how experts play chess. "Expert human players," he writes, "actually seem to acquire a different way of seeing configurations of pieces on the board; they come to experience sophisticated estimates of the possibilities of a position in the very look of the board." But critical to cognitive science is understanding what makes up these representations and how changes in them occur as we progress from beginner to expert.

Computer representation of the world is certainly one of the hardest problems facing cognitive science. Consider a very simple example: What goes into a representation of the act of Mary crossing Elm Street? Does the representation include the speed with which she was walking, the fact that it was raining, the presence of a pickup truck stopped at the light, the type of shoes she was wearing, the *color* of her shoes, the fact that the noon whistle started blowing when she was halfway across the street, the fact that she was wondering how many pounds of berries fermenting in the stomach of an elephant would make it drunk? All of these details *might* be necessary to handle some cognitive situation. The question is: Can situations be represented in a context-independent way (by a representation module), so that all that remains is for some processing engine (e.g., formal logic) to subsequently manipulate them?

All too often, representations are constructed by hand by people who know ahead of time how what these representations need to look like in order to solve the particular problem at hand. Consequently, the human representers leave out many "unnecessary" details and cry victory when their program solves the problem it was designed to solve. But upon closer examination, the battle was already won before the program ever started because the authors of the program chose a representation of the problem that left comparatively little for the program to do.

Finally, there is the question of the book's style. An excessive amount of (frequently non-germane) erudition makes it rather irritating to read. Erudition in the service of content is wonderful; erudition in the service of erudition is distracting and, worse, often hinders clarity. If the relatively new field of cognitive science is to take root, those participating in it must communicate clearly and in a way that transcends the confines of their own special areas.

In summary, those who want to know more about past and present attempts to apply logic to modeling cognition, *An Invitation to Cognitive Science* is certainly worth reading. But for relatively naive readers looking

for an introduction to the field, this is not the best book. This latter audience needs a book that is considerably broader in scope in order to introduce them to the exciting, multidisciplinary endeavor called cognitive science.

[1] Hofstadter, D. H. (1982), Artificial Intelligence: Subcognition as Computation," in F. Machulp and U. Mansfield (eds.), *The Study of Information*. New York: John Wiley.

[2] French, R.M. (1990). Subcognition and the Limits of the Turing Test. *Mind*. Vol. 99, No. 393, 53-65 for a detailed discussion of this point.